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Nano-thick freestanding and reusable epidermal electronics via edge-supporting strategy

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Abstract

Nano-thick epidermal electronics have extraordinary lightweight and conformability characteristics, thus enable high-quality signal acquisition and imperceptible wearing experience. However, reducing the device thickness to nanoscale makes the rigidity extremely low thus the device curls spontaneously, imposing unprecedented challenges to the device fabrication, handling and recycling. To resolve the dilemma of epidermal electronics, we propose an edge-supporting strategy that maintains the benefits of nano thickness, meanwhile obtains practicality and reusability via reinforced edge scaffold. The nano-thick (~500 nm thick) core area of the device provides perfect conformal, firm, and burdenless contact with human skin. On the other hand, the edge scaffold (~5 μm thick) plays an important role in protecting internal fragile film from external forces and providing rigid support during handling and recycling. In addition, the edge scaffold enables spontaneous release of nano-thick electronics from the substrate in water, getting rid of the use of sacrificial layers and etchants. We demonstrate the use of the edge-supporting strategy to develop a physiological electrode with low impedance less than 3.6 k Ω at 1 kHz, a resistance-type temperature sensor with fast response time less than 0.5 ms and a nanoengineering-based pressure sensor with high sensitivity of 62.9 kPa⁻¹. Our work provides an effective method to handle and release nano-thick electronics without using external stamps, thereby enabling truly practical applications of ultrathin flexible electronics in the fields of wearable technology, healthcare and so on.

Introduction

Epidermal electronics are thin and soft electronic devices that can be attached to the human skin for wearable applications such as disease diagnosis and health monitoring^{1–6}. By using soft materials and, more importantly, reducing the thickness of devices, epidermal electronics have a low bending stiffness that ensures their flexibility to adapt to curved surfaces. This characteristic gives epidermal electronics superior performance when compared with conventional and other rigid wearable devices^{7–11}. To date, it has been widely reported that such “skin-like” electronics have fast response, low impedance,

and high signal-to-noise ratio, therefore bypassing limitations of previous approaches¹².

Skin is not only curved but also contains textures with micro three-dimensional structures. Although a large portion of current epidermal electronics can adapt to non-flat surfaces with large curvatures, they are still not able to form conformal contact with micro textures on the skin. According to theoretical analysis by Rogers's group¹³, when the thickness of a polyimide (PI)-based epidermal electronic device reduces to below ~3.8 μm , the device can form fully conformal contact with skin textures, and when the thickness reduces to below ~1.2 μm , mechanically unnoticeable by human skin can be achieved. Furthermore, for tissues with less rigidity such as cerebral cortex, the required thickness of device is expected to be submicron level^{14–16}. Therefore, decreasing the thickness of epidermal electronics to several microns to nanoscale is regarded as the key point to fully fulfill the advantages of extraordinary sensing

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performance and comfortable wearing experience. As a result, many researchers have invested a lot of efforts in this area to promote the development of nano-thick epidermal electronics^{17–21}.

Nevertheless, the decrease in thickness obviously compromises the bending stiffness and structural strength of the nano-thick epidermal electronics, making them easily wrinkled under minor disturbances and difficult to flatten once folded. These issues impose unprecedented difficulties in manipulating nano-thick electronics directly by hand or with normal tools like tweezers, let alone perform some precise operations such as attaching wires or adjusting their position on target surface. Meanwhile, the preparation of these nano-thick epidermal electronics is also challenging. To obtain high quality and intact ultrathin films, it is crucial to minimize the peeling stress during the releasing process of the nano-thick epidermal electronics from their mother substrate. The most commonly used method is to introduce a sacrificial layer^{22–25}. Once the sacrificial layer is dissolved in an etchant, the device can be released without suffering from interface competitive fracture caused by peeling. For example, Konagai et al.²⁶ introduced an AlGaAs sacrificial layer between GaAs functional layer and the silicon substrate. By utilizing the faster etching rate of the AlGaAs in the hydrofluoric acid etchant compared to that of the GaAs, the functional structural layer was priority released from the surface. To get rid of the using of chemical etchants, Wu et al.²⁷ introduced a type of water-soluble sacrificial layer using ion-crosslinked polyacrylic acid between the gold thin-film electrode and the silicon substrate. However, after dissolving this sacrificial layer in water, the gold thin-film electrode is not self-supporting thus needs to be transferred onto a hydrogel substrate for temporary support. This makes the device not re-useable and transferring the device from the hydrogel substrate to the target induces excessive stress. In addition, the sacrificial layer present challenges in terms of process compatibility²⁸. Meanwhile, residual sacrificial material may remain on the transferred functional layer or the donor substrate, and further affect the performance of thin films and the recovery and reuse of donor substrates. Besides sacrificial layers, researchers have also developed several other methods to separate nano-thick epidermal electronics from the mother substrate, such as thermal-assisted releasing^{29,30}, laser-assisted peeling^{31,32}, and controlled spalling^{33,34}. However, these strategies cannot completely avoid interface competitive fracture, therefore always leaving a risk of rupture during the peeling process. In addition, transfer printing methods can transfer ultrathin devices onto the skin or other complex surfaces^{35–39}, while they neglecting the recycling of ultrathin devices without damage, which is crucial for the reuse and long-term performance investigation of devices.

To resolve the conflict between the high performance and practicality of nano-thick epidermal electronics, here we propose an innovative edge-supporting strategy, in which the internal sensitive area is reduced to nano-thick (~ 500 nm), while the peripheral edge of the device is reinforced to the micron scale (~ 5 μ m) as the supporting scaffold. The nano-thick core enables perfect conformal contact with human skin, providing exceptional sensing performance. The edge scaffold protects internal fragile film from tearing during peeling from the substrate and provides rigid support during handling. Also, the edge scaffold can release from the rigid substrate by simply immersing it in water for several minutes and initiates the spontaneous peel-off of the whole nano-thick epidermal electronics without using sacrificial layers, etchants, or transfer stamps. In addition, with the support of edge scaffold, the nano-thick epidermal electronics can detach from the target surfaces and re-attach for repeatedly use. Compared to previous studies, our strategy offers an effective and reliable solution to handle, fabricate and reuse nano-thick epidermal electronics, thereby greatly facilitating the practical application of these devices. To demonstrate the practical use of this strategy, we developed a kind of physiological electrodes with contact impedance less than 3.6 k Ω at 1 kHz (on human skin) and long-term drift (after 1-week) less than 120%, which are three orders of magnitude and more than fourfold better than commercial electrodes (2.1 M Ω and 568% respectively). We also developed an ultrathin resistance temperature detector (RTD) with response time less than 0.5 ms and a pressure sensor with sensitivity of 62.9 kPa⁻¹. The thickness of sensitive area of both designs is less than 1 μ m. Our strategy addresses a key bottleneck of nano-thick epidermal electronics, achieving extraordinary performance and operability, thus demonstrating great potential to benefit the community.

Results and discussion

Design and fabrication of nano-thick epidermal electronics via edge-supporting strategy

Figure 1 explains the edge-supporting strategy and its advantages in fabricating and handling nano-thick epidermal electronics. A typical design is illustrated as Fig. 1a. The core area of the device is made of a nano-thick film with sensitive unit (e.g. electrode, sensor). The thickness of the sensitive unit is 500 nm (as illustrated by the inserted figure in Fig. 1a). To support the nano-thick film, the edge of the core area is thickened to ~ 5 μ m, serving as supporting scaffold to prevent excessive concentrated stress from damaging the fragile ultrathin film. Considering that the tensile stress will mainly concentrate at the sharp corners, the thickened edge is designed as the most universal circular shape. The circular design eliminates sharp corners where stress typically localizes,

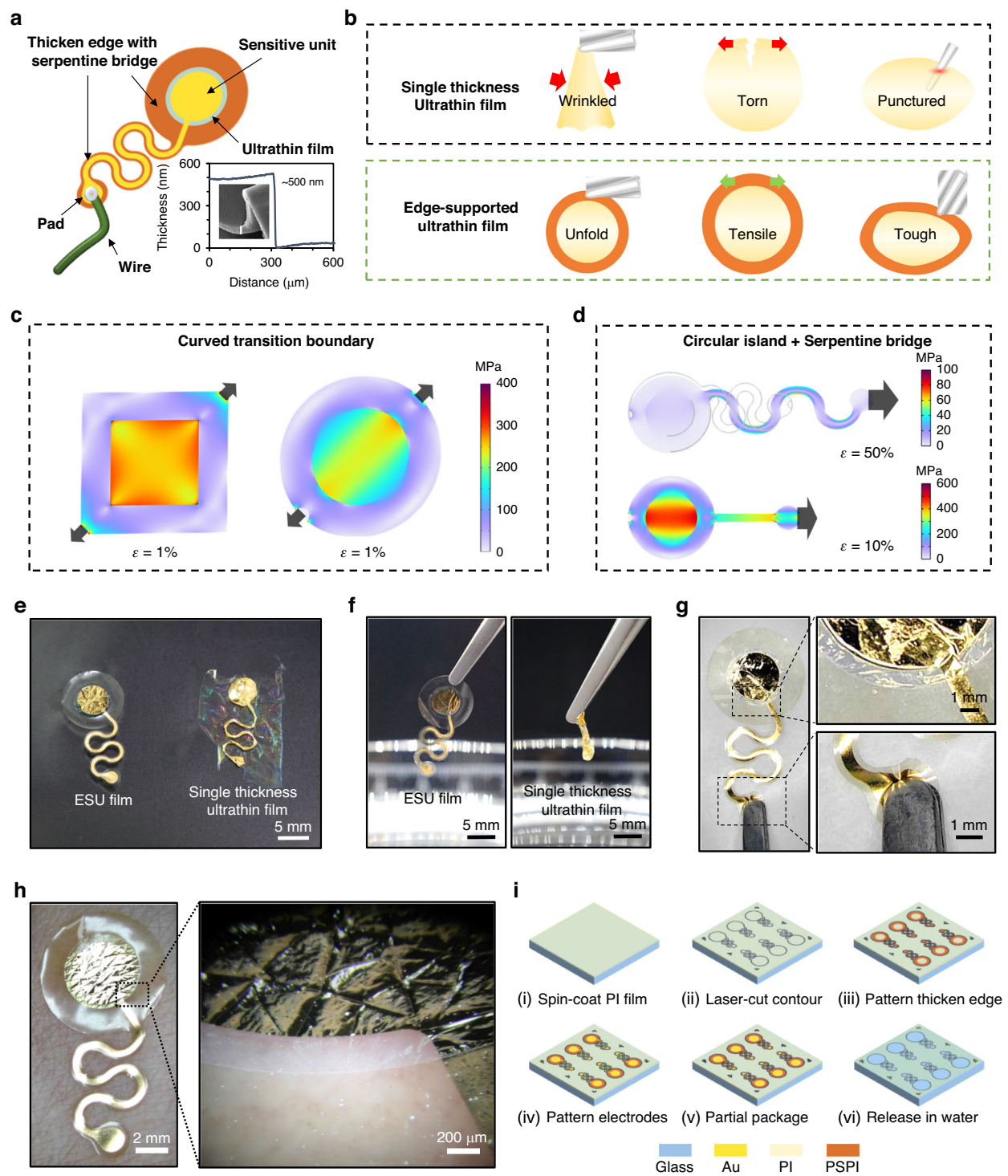


Fig. 1 Edge-supporting strategy for nano-thick epidermal electronics. **a** Schematic diagram of typical edge-supported ultrathin epidermal electronics. **b** Comparison of nano-thick epidermal electronics with and without reinforced edge in terms of operability. Structural optimization of ESU film at **c** circular shape sensing area and **d** serpentine shape leading wire. **e** Handling of the nano-thick devices when carried in aqueous solution and **f** picked up by tweezers in air. ESU is short for edge-supported ultrathin. **g** The functional differentiation design strategy: ultrathin film for sensing, thickened edge for handling and serpentine bridge for stretchability. **h** Demonstration of conformal contact with human skin. **i** Fabrication process of the edge-supported nano-thick epidermal electronics. PSPI is short for photosensitive polyimide

thereby providing isotropic mechanical robustness regardless of the stretching direction. This geometric advantage is particularly critical for thin-film devices that may undergo unpredictable or multi-directional deformation during handling or operation. In contrast, other regular polygons (e.g., square, hexagonal) or asymmetric shapes inevitably introduce corners or anisotropy, which can act as stress concentration sites under specific loading conditions. The simulation results was shown in Fig. 1c, the curved transition boundary can improve the distribution of stress and relatively reduce the maximum stress $\sigma_{\max}$ from 261 MPa to 121 MPa at tensile strain ε of 1%, which is below the tensile strength of PI (~ 200 MPa). Besides the edge scaffold, a serpentine island-bridge structure with a thickness of $\sim 5 \mu\text{m}$ is connected to the circular device. In this case, the solder pad is thickened to withstand the stress during soldering and pulling of the wire. In addition, the serpentine structure further enhances the stretchability and reduces stress distributed in island areas through its bending and twisting deformation. As shown in Fig. 1d, the structure without the serpentine design exhibits a $\sigma_{\max}$ of 570 MPa at $\varepsilon = 10\%$. In contrast, the device with the serpentine structure shows a $\sigma_{\max}$ of 79.9 MPa at $\varepsilon = 50\%$, and the stress in the nano-thick ultrathin island area is only 7.47 MPa. Using this strategy, the internal PI film carrying the sensitive unit was ultrathin to obtain high performance, while the edge scaffold and solder pad area provide sufficient rigidity to support the device and enable deformation as well as wire bonding. Compared to other reported works, the edge-supported ultrathin devices exhibit significant advantages in terms of excellent operability, as shown in Fig. 1b. Conventional ultrathin flexible electronics possess extremely thin thickness and lightweight quality, making them highly susceptible to small disturbances such as intermolecular forces, electrostatic forces, and airflow during manipulation, resulting in unpredictable deformations. The unexpected wrinkles and folds prevent these ultrathin films from accurately and flatly covering the target area, despite their excellent conformal capability. In contrast, the thickened edge provides sufficient strength to withstand pressure, stretching, and twisting (Fig. 1b). As demonstrated in Fig. 1e-h, the edge scaffold is able to support the whole device without wrinkles in water or air (Fig. 1e, f) during fabrication, and protect fragile ultrathin film from tearing at particular boundary or withstand operations such as clamping or wiring (Fig. 1g). Meanwhile, the nano-thick core area can form conformal contact with human skin with micro-texture duplicated obviously, demonstrating negligible impact caused by the reinforced edge (Fig. 1h).

Figure 1i illustrates the fabrication process of a typical edge-supported nano-thick epidermal electronic device. First, a layer of nano-thick PI (~ 500 nm) is spin-coated

onto a glass substrate and then thermally cured. PI was selected as the flexible substrate due to its excellent biocompatibility, chemical stability and high dielectric constant. Then, the unwanted regions are removed via laser-cutting, leaving behind the nano-thick core area along with the defined peripheral edge on the substrate (Supporting Information, Fig. S1). After that, a layer of photosensitive PI ($\sim 5 \mu\text{m}$) is spin-coated and patterned to form the thickened edge scaffold. Subsequently, a layer of functional materials (e.g. Au, 50 nm) is patterned on the PI film. Finally, the fabricated edge-supported nano-thick devices are spontaneously released after sequential rinsing in ethanol and deionized water in few minutes. The relationship between the final PI film thickness and spin-coating speed is presented in Fig. S2 (Supporting information), and more details about the fabrication process can be found in “Methods” section.

Water-releasing mechanism for edge-supported nano-thick epidermal electronics

Detachment of the fabricated epidermal electronics from the mother substrate is a critical step for manufacturing, while it is difficult to peel off such a fragile film from the mother substrate in a damage-free and intact manner. Whether the film can be peeled off from the substrate essentially depends on the competition between the force acting on the film and the adhesion between the film and mother substrate. Existing methods that rely on external forces are vulnerable to defects and incomplete detachment. The edge-supporting strategy provides an effective way to enable the spontaneous release of nano-thick epidermal electronics from substrate in aqueous solution without requiring external forces or chemical etchants. The underlying mechanism and experimental results are presented in Fig. 2. The thickened edge not only protects the internal nano-thick film from tearing during the releasing process but also provides an internal driving force to initiate the nano-thick film's detachment from the mother substrate. The imidization temperature during the PI curing process can reach over 200°C , while the coefficients of thermal expansion of PI and glass substrate are significantly different with an order of magnitude, which causes an accumulation of thermal mismatch stress when the PI film is cooled. As shown in Fig. 2a, the finite element analysis results describe the distribution of the thermal mismatch stress when the thickness of PI film is different. Since the contact surface between PI film and mother substrate is constrained while the upper surface exposed in air is free, the PI film as a whole tends to curl upwards under the action of thermal mismatch stress. Moreover, this trend increases with film thickness and will trigger an avalanche-like detachment once a local detachment occurs. As shown in Fig. 2b, the deflection of glass substrate ($50 \text{ mm} \times 50 \text{ mm} \times 1 \text{ mm}$)

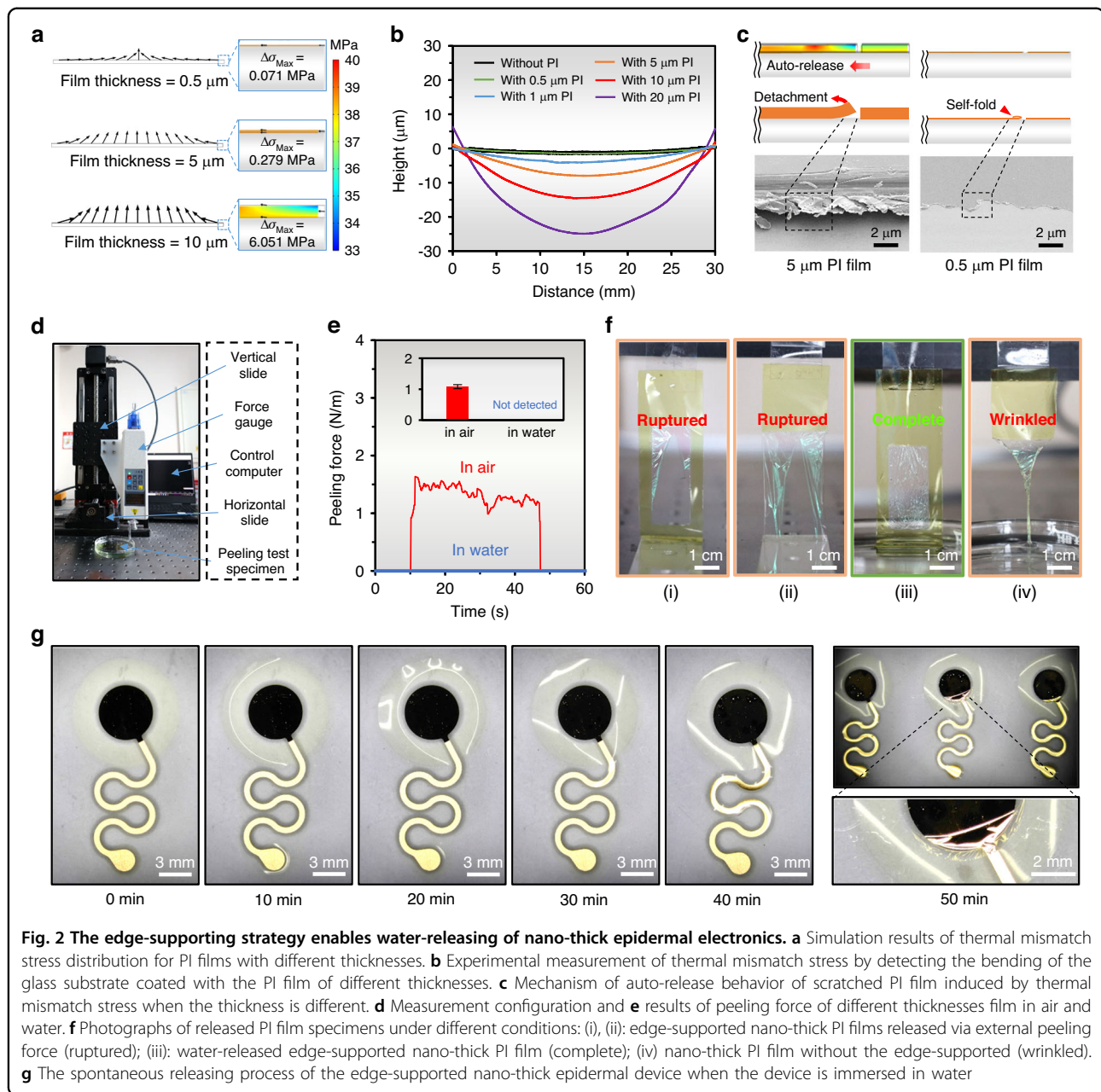


Fig. 2 The edge-supporting strategy enables water-releasing of nano-thick epidermal electronics. **a** Simulation results of thermal mismatch stress distribution for PI films with different thicknesses. **b** Experimental measurement of thermal mismatch stress by detecting the bending of the glass substrate coated with the PI film of different thicknesses. **c** Mechanism of auto-release behavior of scratched PI film induced by thermal mismatch stress when the thickness is different. **d** Measurement configuration and **e** results of peeling force of different thicknesses film in air and water. **f** Photographs of released PI film specimens under different conditions: (i), (ii); edge-supported nano-thick PI films released via external peeling force (ruptured); (iii); water-released edge-supported nano-thick PI film (complete); (iv) nano-thick PI film without the edge-supported (wrinkled). **g** The spontaneous releasing process of the edge-supported nano-thick epidermal device when the device is immersed in water

covered with thermal-cured PI film of different thicknesses was measured by step profiler (ET4000M, KOSAKA, Japan), from which it can be found that the deflection of glass substrate increased after covering with a thicker PI film. As described in Fig. 2c, a common approach to initiate this phenomenon is to cut off the edge of the film. The maximum stress gradient moves with connection edge between the unreleased film and mother substrate, and detaches the film from the mother substrate when the stress gradient is greater than adhesion force between film and mother substrate. When using a surgical blade to cut the PI film, the tip of blade

provides a peeling force to preliminarily lift the edge of film up from the mother substrate (Supporting information, Fig. S3 and Movie S1). Therefore, the cutting edge of the 5 μm thick PI film can create an initial gap for detachment to propagate from the substrate, forming a gap for further development of detachment. For comparison, the thermal mismatch stress of submicron thickness PI film is too weak to peel it off from the mother substrate even after the film was cut. Because of the high flexibility of the submicron thickness PI film, the cut edge becomes curled and folded to suppress the development of detachment. While the 5 μm thick PI film provides an

initial driving force for detachment, the cascading release of adjacent areas cannot propagate because the interfacial adhesion force remains dominant. Therefore, reducing the adhesion force between the film and the substrate also needs to be conducted. It is found that water environment can effectively reduce the adhesion force between film and its mother substrate. As the highly polar water molecules gradually infiltrate the interface between the PI film and substrate, the bonding force between the two weakens. We measured the adhesion force of submicron thickness PI film in air and in water. The details of adhesion force test can be found in “Methods” section. As shown in Fig. 2d, e, the adhesion force recorded by a force gauge was 1.08 ± 0.06 N/m in air, whereas no measurable adhesion was detected in water. Consequently, once an initial gap forms between the PI film and the substrate, water molecules can invade the interface and promote the complete detachment of the entire film.

In summary, by adding the 5 μm thick edge scaffold and immersing the device in the water, the nano-thick epidermal electronic device can be released from the substrate spontaneously and non-invasively. Figure 2f presents images of different specimens. The edge-supported nano-thick devices that peeled off in the air eventually become ruptured and damaged (Fig. 2f(i) and f(ii)), whereas the device released in water is intact with no defects (Fig. 2f(iii)). Besides, the nano-thick device without the edge-supported becomes severely wrinkled (Fig. 2f(iv)). Figure 2g shows the water-releasing process of the edge-supported nano-thick epidermal device in water. In contrast, an ultrathin edge can prevent the internal thick area from detaching (Supporting Information, Fig. S4). Furthermore, the thickened edge and serpentine island-bridge structure significantly reduce the risk of damage to the internal ultrathin film when stretched during operation (Supporting Information, Fig. S5). These above-mentioned features enable the ESU film to be reliably, completely, and facily peeled off from the mother substrate, which is crucial for the yield of flexible electronic manufacturing.

Conformal contact ability of the edge-supported nano-thick epidermal electronics

The high performance of ultrathin flexible electronics mainly comes from their perfect conformability. According to theoretical analysis in previous works^{13,39}, a smaller device thickness is crucial for flexible electronics to achieve conformal contact with the details of target surface. Figure 3a presents the calculation results of the critical thickness at which a PI film (with 50 nm thick Au electrode on it) can achieve conformal contact with human skin. For instance, when the roughness amplitude of the fingerprint is about 200 μm , to achieve conformal contact it requires the PI film thickness to be below

~ 3.5 μm for this specific texture. Moreover, the 0.5 μm thick PI film can easily conform to the surface with roughness amplitude ranging from 60 to 500 μm (encompassing various skin undulations) and beyond. As shown in Fig. 3b, the submicron thickness PI film (with thickness of 0.5 μm) can easily form conformal contact with the replica of fingertip. The surface topography profile of the bare fingerprint and the fingerprint covered by PI films of different thicknesses was measured by step profiler and compared in Fig. 3c. The PI film with thickness of 0.5 μm fully adheres to the undulations of fingerprints, inducing almost no compression deformation on the morphology of fingerprints, which is important for comfortable wearing. Conversely, the 5 μm thick PI film achieves only partial contact, primarily by compressing the ridge tops. Figure 3d, e shows the scanning electron microscopy (SEM) images of fingertip replicas covered by 5 μm and 0.5 μm PI film, respectively. The images clearly demonstrate that the fine morphology around sweat pores remains visible through the 0.5 μm film, as observed from both side-view SEM (Fig. 3e) and top-view optical microscopy (Fig. 3f). On the other hand, due to the poor air and water permeability of PI film, prolonged adhesion of the film to the surface of human skin will cause significant redness without swelling, and slight discomfort (Fig. S6). This is also one of the important reasons why the electrode is designed to be removable in this work.

Next, we investigated the conformal contact capability of the edge-supported nano-thick epidermal device, using the design shown in Fig. 1f as an example. Figure 3g presents the surface topography of a human forearm skin replica covered by the device. The thickened edge and the part of nano-thick film that is nearby and connected to it have incomplete contact with the skin, but this does not affect the signal acquisition quality of the sensing electrode. The sensing area (500 nm thick) is marked by a red circle with diameter of 5 mm. The majority of nano-thick sensing area achieves full conformal contact with the skin, despite its complex morphology (more intricate than a fingerprint). By analyzing the data of the contact line in the diameter direction, it can be estimated that the proportion of effective contact line exceeds 85%. As conclusion, the influence introduced by the height difference on the conformal contact can be completely avoided at the sensing area through reasonable size design. A more detailed discussions related to the design of the thickness and width of the thickened edge are presented in the Supplementary information.

Edge-supported nano-thick epidermal electronics for high-quality electromyography acquisition

Surface electromyography (sEMG) signals are muscle electrical activities obtained via electrodes placed on the

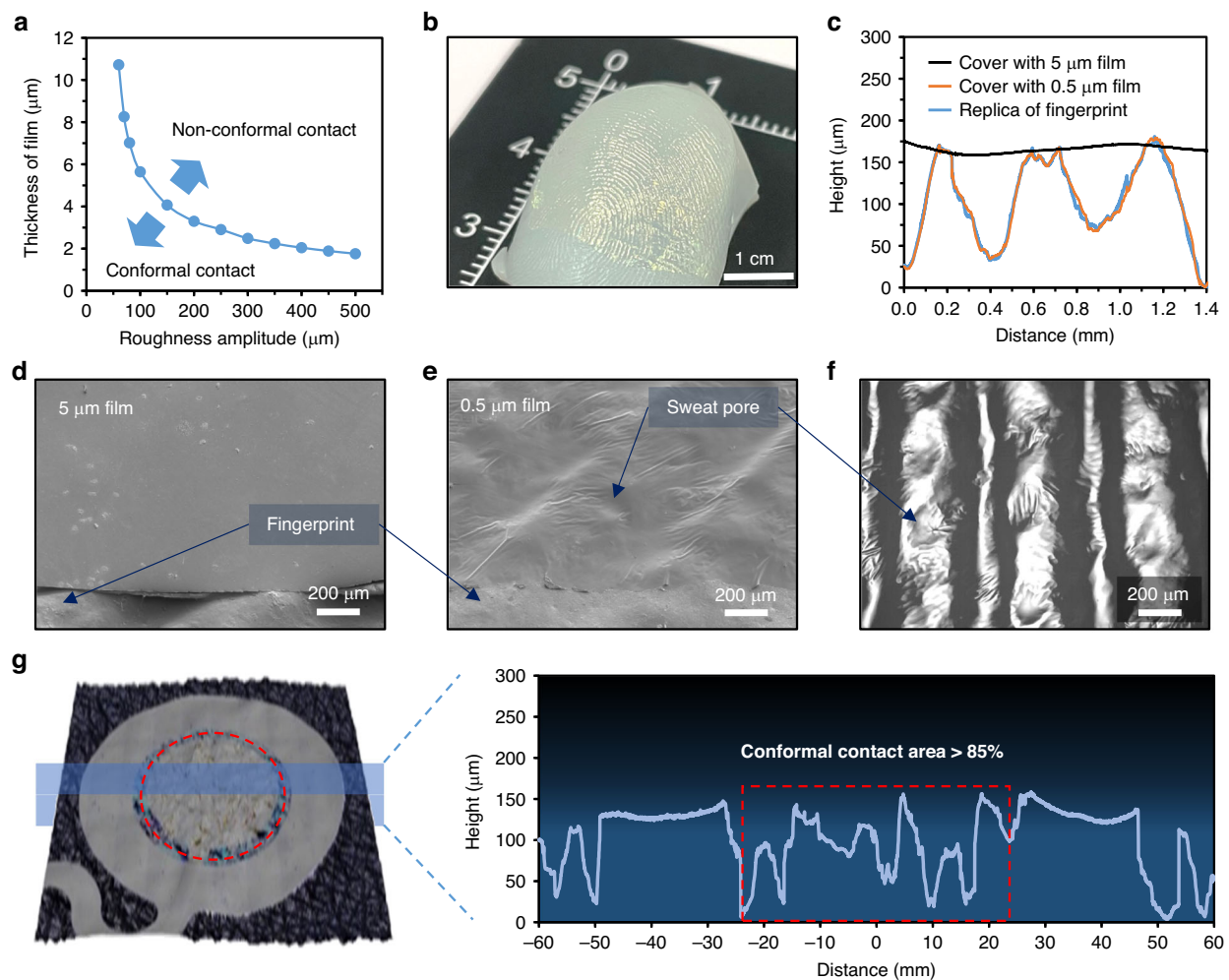


Fig. 3 Conformal contact capability of edge-supported nano-thick epidermal electronics. **a** Critical film thickness required for achieving conformal contact as a function of surface roughness. **b** Photograph of a submicron-thick PI film conformally attached to a fingertip replica. **c** Surface topography profiles and SEM image of the fingertip replica covered by PI films with thicknesses of **d** 5 μm and **e** 0.5 μm . **f** Optical image of the replica of fingertip covered by a 0.5 μm thick PI film. **g** Surface topography of the edge-supported nano-thick device attached to a human forearm skin replica, captured by confocal microscopy. The red circle and the red box mark the 5 mm diameter sensing area and the conformal contact area, respectively

skin. They are used to reflect the degree of muscle activation, fatigue status, and movement control mode. High-quality sEMG acquisition requires conformal electrode-skin contact to minimize interfacial impedance. Commercially prevalent Ag/AgCl wet electrodes employ conductive gel to enhance adhesion and contact quality, offering ease of use and high-fidelity signals. However, their drawbacks include potential skin irritation and signal degradation over time as the gel dries. The dry electrodes based on ultrathin film are supposed to be an alternative replacement for commercial wet electrodes, as their perfect conformal contact ability provides a larger effective contact area and lower contact impedance. Nevertheless, their practical application is hampered by poor mechanical robustness and challenging handling.

The edge-supported design presented here represents a structurally optimized ultrathin dry electrode that addresses these key limitations. Figure 4 shows the design and characterization results of this novel sEMG electrode based on the edge-supported nano-thick epidermal platform. As shown in Fig. 4a, to match the connectors of commercial lead wires, the electrode pads were designed as a ring shape with the inner and outer diameter of 4 mm and 8 mm, respectively. As shown in Fig. 1b(i), the circular welding pad of our ESU electrode can be embedded into a standard electrode buckle for direct connection to a signal acquisition device via a commercial lead wire. As a result, the ESU electrode can be repeatedly attached or lifted to be placed at the appropriate measurement position (Movie S2). Once the ESU electrode approaches and

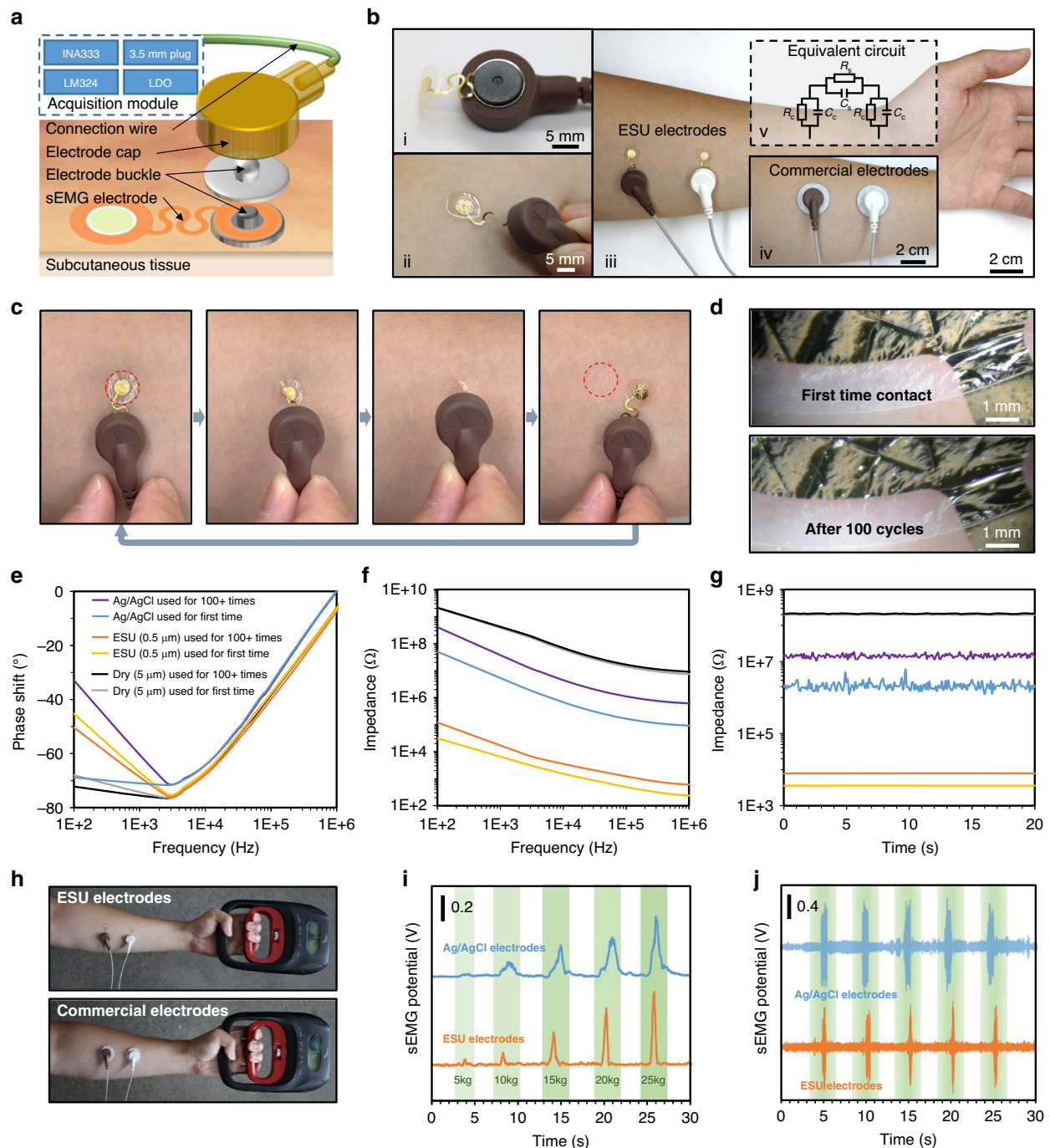


Fig. 4 ESU electrodes with low impedance and long-term stability. **a** Schematic diagram of ESU electrodes integrated with commercial lead. **b** Configuration of electrical characterization of ESU electrodes, thick dry electrodes and commercial wet electrodes. **c** The process of removing the device without damage. **d** The interfacial integrity of the ESU after repeated on-skin application. **e** Phase shift and **f** impedance magnitude spectra (100 Hz to 1 MHz) of the three electrode types. **g** Impedance magnitude at a constant frequency of 1 kHz. Blue, yellow, and gray lines represent the results of wet electrodes, ESU electrodes and thick dry electrodes, respectively, and violet, orange, and black lines represent the results of these electrodes after 1-week of measurement, the details of test can be found in "Experimental section". **h** Configuration of electrodes recording sEMG signal on human forearm. sEMG signals recorded during gripping tasks: **i** under varying grip strength, and **j** during repeated grips of constant strength. All electrical measurements were conducted at 1 kHz

comes into contact with the target surface, the sensing electrode on ultrathin film will adhere tightly to the target surface by van der Waals forces (Fig. 4b(ii)). Figure 4b(iii) illustrates the sEMG measurement setup: two ESU electrodes were attached to the forearm skin to form a bipolar lead for recording sEMG signals during flexor carpi radialis contraction. For comparison, a pair of commercial Ag/AgCl electrodes (Fig. 4b(iv)) and a pair of thick dry electrodes (with an identical design but a sensing area thickness of 5 μm) were placed at the same location. The equivalent circuit of the measurement lead is shown in Fig. 4b(v). The impedance during sEMG signals acquisition is primarily determined by the electrode-skin interface (modeled by R_c and C_c), and the subcutaneous tissue (modeled by R_s and C_s).

Figure 4c demonstrates the process of removing the device without damage, the embedded ESU can be easily removed from the skin and leaves no residue. Figure 4d exhibits the interfacial integrity of the ESU after repeated on-skin application, the transition between the 5 μm scaffold and the 0.5 μm electrode area shows no delamination or fatigue cracking over 100 cycles of conformal contact to skin and subsequent removal. It is worth mentioning that small pieces of dead skin cells or sebum will inevitably adhere and accumulate on the surface of ESU electrodes over repeated uses, especially the on the surfaces of nano-thick core area. Therefore, before and after each attachment, the electrode needs to be rinsed in ethanol for few seconds, and the target skin area should be gently wiped with an alcohol cotton ball to minimize surface contamination that may affect electrode performance^{40–42}, such as surface adhesion and contact impedance. The contact impedance amplitude and phase shift of the edge-supported nano-thick electrodes were compared with those of commercial wet electrodes and thick dry electrodes (Fig. 4e–g). Compared to the thick dry electrodes, the phase shift of the edge-supported nano-thick electrode more closely approximates that of the commercial wet electrodes (Fig. 4e), which is desirable as a phase angle near 0° minimizes signal distortion. After one week of use, the phase shift of the wet electrodes degraded, approaching the poorer performance of the thick dry electrodes, whereas the edge-supported electrodes exhibited negligible change. As shown in Fig. 4f, reducing the dry electrode thickness from 5 μm (gray line) to the submicron level (0.5 μm , yellow line for edge-supported) yielded a contact impedance lower than that of commercial Ag/AgCl electrodes (blue line) across the frequency range of 100 Hz to 1 MHz. As shown in Fig. 4g, at 1 kHz, a frequency associated with lower noise and suitable for sEMG signal acquisition, the measured impedances were $3.60 \pm 0.01 \text{ k}\Omega$ for the edge-supported nano-thick electrode, $2.15 \pm 0.57 \text{ M}\Omega$ for the commercial Ag/AgCl electrode, and $198.27 \pm 1.36 \text{ M}\Omega$ for the thick dry electrode.

To assess long-term performance, the impedance variation of electrodes was recorded and marked with asterisks in Fig. 4e–g. As shown in Fig. 4e, after more than 100 repeated detachments and signal acquisitions over the course of one week, the impedance of the edge-supported nano-thick electrodes at 1 kHz was $7.908 \pm 0.006 \text{ k}\Omega$. In contrast, the impedance of the Ag/AgCl electrodes increased by over an order of magnitude to $14.353 \pm 1.564 \text{ M}\Omega$. Furthermore, the relative signal fluctuation of the edge-supported nano-thick electrodes (0.11%) was significantly smaller than that of Ag/AgCl electrodes (26.64%). The pronounced impedance drift in wet electrodes is mainly attributed to the deterioration of the hydrogel over time and with repeated use. While dry electrodes avoid this issue, they may experience a minor impedance increase due to surface contamination during handling. An on-body test was performed as illustrated in Fig. 4h. The edge-supported nano-thick electrodes were attached to the inner forearm to record the sEMG signals generated during a gripping action. As shown in Fig. 4i, the signal amplitude from the edge-supported electrodes increased with grip strength in a manner similar to, and even more rapidly and sensitively than that of the wet electrodes. Figure 4j compares the signal quality. For the edge-supported electrode, the standard deviation, noise, and signal-to-noise ratio were 240.75 mV, 29.44 mV, and 19.92 dB, respectively. The corresponding values for the wet electrode were 240.03 mV, 40.46 mV, and 15.46 dB, respectively. These results indicate that the edge-supported nano-thick electrodes offer performance comparable to commercial Ag/AgCl wet electrodes, with superior stability for long-term applications.

Edge-supported nano-thick epidermal electronics for high-quality temperature and pressure sensing

Beyond electrophysiological sensing, temperature and pressure monitoring represent another crucial application domain for epidermal electronics. The edge-supporting strategy is pivotal in developing highly precise and sensitive sensors for these parameters. As shown in Fig. 5a, we demonstrate a kind of RTD and a kind of pressure sensor based on nanoengineering design. The thin-film resistive type RTDs have very stable linear response and a simple structure. As the substrate thickness of RTD decreases, its temperature coefficient of resistance (TCR) tends to be improved (Fig. 5b) and close to the intrinsic value of its sensitive material (Au, 0.324%). This is primarily because, for a polymer substrate with poor thermal conductivity, a thinner substrate reduces the temperature difference between the temperature sensing unit and the heat source that is caused by the substrate itself (Fig. S7). More significantly, the response time improves dramatically because the thermal mass of the

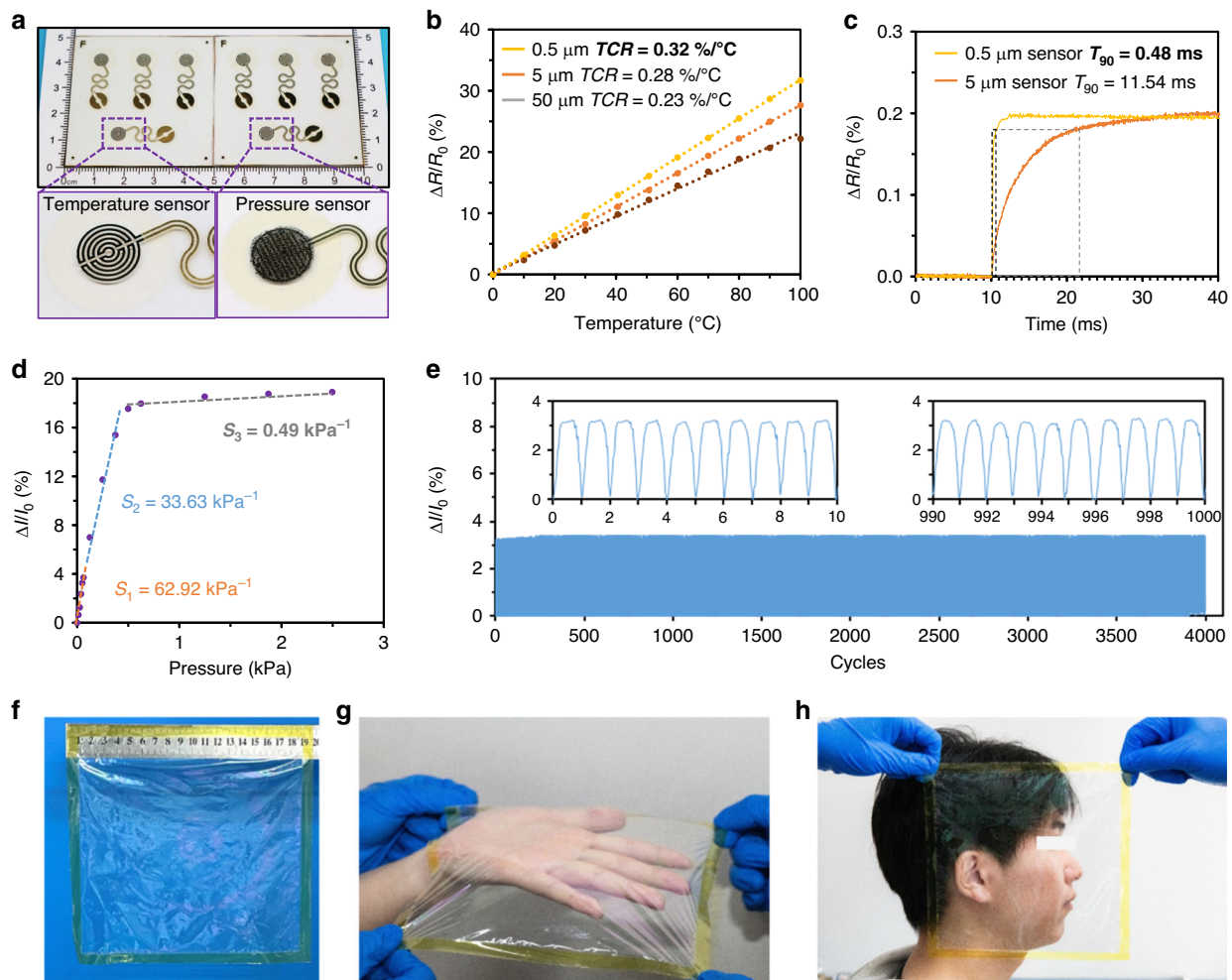


Fig. 5 Performance of ESU sensors and prospects for large-area applications. **a** Schematic illustrations of the developed ESU devices: a resistive temperature detector (RTD) and a pressure sensor. **b** Resistance-temperature relationship and **c** response time of RTDs with different substrate thicknesses. **d** Current-pressure response curve and **e** cycling durability test of the ESU pressure sensor. **f-h** Demonstration of the ESU film as a large-area, flexible substrate for multifunctional electronic skin, highlighting its potential for scalable wearable applications

encapsulation layer between the sensing unit and the heat source decreases with thickness⁴³. As shown in Fig. 5c, the response time of a 5 μm -thick RTD is 11.54 ms and is improved to 0.48 ms when the thickness of RTD is 0.5 μm . This study successfully promotes the response time of metal-based RTDs to sub-millisecond level for the first time⁴⁴.

For a long time, the conventional strategy to achieve a high sensitivity of resistive type pressure sensor is introducing microstructure in its active layer^{18,45–47}. Stress concentration at the contact points of microstructures causes a large change in contact area and leads to a significant resistance variation^{46,48,49}. However, the different design of microstructures shares a common drawback that inevitably increases the thickness of sensors. In order to achieve both ultrathin thickness and high sensitivity simultaneously, a novel

nanoengineering strategy was proposed for pressure sensor design in our previous work^{39,50}. As demonstrated in Fig. 5d, e, the edge-supported nano-engineered pressure sensor exhibits high sensitivity (62.92 kPa^{-1} in its most sensitive range) and excellent durability, with the latter owing to the thickened edges which aid post-deformation recovery. In addition to the present examples, ESU films also show broad potential for applications requiring large-area flexible substrates (Fig. 5f-h).

Conclusions

In summary, this paper demonstrates an edge-supporting strategy—a novel design and fabrication approach—for developing nano-thick epidermal electronics. By locally tailoring the film thickness, distinct functionalities are integrated. The internal submicron

thickness region ensures intimate, conformal contact between the sensing unit and human skin, enabling high device performance. Meanwhile, the external thickened region serves as a reinforced scaffold, not only protecting the fragile internal film from tearing but also providing mechanical support to prevent wrinkling during fabrication and operation. Furthermore, the thickened edge facilitates the water-released process of the whole device from the mother substrate, eliminating peeling stress and the need for external transfer stamps. Using this strategy, we successfully fabricated three representative epidermal devices, including a physiological electrode with low impedance less than $3.6\text{ k}\Omega$ at 1 kHz, a resistance-type temperature sensor with fast response time less than 0.5 ms, and a nanoengineering-based pressure sensor with high sensitivity of 62.9 kPa^{-1} . This work provides a reliable and damage-free method for fabricating and handling nano-thick epidermal electronics, effectively overcoming a key obstacle to the practical application of ultrathin devices.

Experimental Section

Consent to participate

Informed consent has been obtained from the volunteers who participated in the experiments.

Fabrication process of edge-supported nano-thick film with serpentine island-bridge structure

First, a PI precursor solution (PAA-1003, Changzhou Yaan New Materials, China) was diluted with NMP solution (Changzhou Yaan New Materials, China) with a volume ratio of 1:2. Then, the mixture was stirred for 30 min and then placed in vacuum for 30 min to remove bubbles. Next, the diluted PI precursor solution was spin-coated on a $50\text{ mm} \times 50\text{ mm}$ glass sheet at 600 rpm for 9 s and 2000 rpm for 30 s, and then was step-cured in an oven from 60°C to 220°C with a temperature gradient of 20°C and a duration of 10 min at each stage. Subsequently, the releasing edge was cut on the formed ultrathin PI film ($\sim 0.5\text{ }\mu\text{m}$) by a laser cutting machine (ProtoLaser R4, LPKF Laser & Electronics, Germany) with the laser power of 0.1 W and frequency of 100 kHz. After that, a PI photoresist (BL301, Asahi Kasei, Japan) was spin-coated on ultrathin PI film at 600 rpm for 9 s and 2000 rpm for 30 s, and prebaked on a hot plate at 110°C for 4 min. Next, the PI photoresist layer was exposed for $200\text{ mJ}/\text{cm}^2$ (MA/BA6, SUSS Micro Tec, Germany) after being covered with a film mask (Shenzhen Qingyi Photomask, China), and then patterned by developer (RS120, Resemi, China) after post-baking on a hot plate at 110°C for 15 min. At last, the patterned PI photoresist layer was step-baked using the same step-curing procedure as that for the ultrathin PI film.

Fabrication of edge-supported nano-thick sEMG electrode and temperature sensor

First, a 50 nm thick Au layer was deposited on the fabricated edge-supported nano-thick PI film by magnetron sputtering (Lab 18, Kurt J. Lesker, USA). Then, a positive photoresist layer (AZ MIR 703, AZ Electronic Materials, USA) was spin-coated on Au layer at 600 rpm for 9 s and 2500 rpm for 30 s, and soft baked on an 85°C hot plate for 30 min. After that, the photoresist layer was exposed for $350\text{ mJ}/\text{cm}^2$ (MA/BA6, SUSS Micro Tec, Germany) after covering with a film mask (Shenzhen Qingyi Photomask, China), and patterned in a developer (AZ 400 K, AZ Electronic Materials, USA) to serve as a mask for the wet etching of Au film. Subsequently, the Au layer was immersed in mixture of KI, I_2 and H_2O (10 g: 2 g: 100 mL) for 30 s to remove the region without protection of patterned photoresist mask. Finally, the photoresist layer was flood-exposed to $350\text{ mJ}/\text{cm}^2$ and then removed by developer. After above-mentioned process, the fabricated device was released in water.

Fabrication of edge-supported nano-thick pressure sensor

First, an edge-supported nano-thick film with patterned Au interdigital electrode on it was fabricated by the previous process. Next, a polyacrylic acid (PAA) solution (25% in water, Polysciences Inc., USA) was diluted with deionized water with a volume ratio of 1:10 and stirred for 30 min and then placed in vacuum for 30 min to remove bubbles. After that, the diluted PAA solution, which serves as the water solvable sacrificial layer, was spin-coated on a $50\text{ mm} \times 50\text{ mm}$ glass sheet at 600 rpm for 9 s and 2000 rpm for 30 s, and then cured on a hot plate at 85°C for 30 min. Subsequently, a polydimethylsiloxane (PDMS) (SYLGARD 184, Dow Chemical Company, USA) mixture (precursor and curing agent) with a weight ratio of 10:1 was diluted with toluene (Meryer (Shanghai) Biochemical Technology, China) and placed in vacuum for 30 min to remove bubbles. After that, the diluted PDMS solution was spin-coated on PAA sacrificial layer at 600 rpm for 9 s and 4000 rpm for 30 s, and then cured on a hot plate at 85°C for 30 min. At last, the single-walled carbon nanotube solution ($150\text{ }\mu\text{g}/\text{mL}$, XFNANO Materials Tech Co., Ltd) was spray-coated on the PDMS film through a stencil mask.

Water-releasing process of the ESU devices

The ESU devices were immersed in DI water until most part of thickened edge departs from substrate, which took for 10–30 min depends on the size and design. Gently stirring the DI water can accelerate the separation of the thickened edge from substrate. Then, the devices could be lifted by clamping the supporting edge using tweezers to completely detach from the substrate and remove from water.

Fabrication process of fingertip/skin replica

First, a mixture of silicone rubber (Smooth-On, USA) was poured into a square groove without a cover. Then the finger/skin was pressed gently into the semi-cured rubber and held until the rubber was completely cured at room temperature. After that, the rubber mold was slowly peeled off from finger/skin, and a layer of release agent (Ease Release 200, Smooth-On, USA) was sprayed on the surface of the rubber mold. Next, a mixture of silicone rubber was poured into the rubber mold and cured at room temperature. Finally, the replica of fingertip/skin was peeled off carefully from the rubber mold.

Characterization of edge-supported nano-thick electrode, temperature sensor and pressure sensor

The contact impedance of a physiological electrode was measured by a precision LCR meter (TH2840B, Tonghui, China). The impedance testing of the dry electrodes, wet electrodes, and ESU electrodes lasted for one week. Each day during the test, the electrodes were first attached to the skin surface to acquire and record one set of signals. Then, after repeatedly attaching and detaching the electrodes to the skin surface twenty times, the electrodes were again attached to the skin surface to acquire and record another set of signals. The resistance of the temperature sensor and the pressure sensor was measured by an eight-channel digital multimeter (TRION-2402-MULT-8-L0B, Dewetron, Austria).

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Author contributions

Z.L., S.W., C.W., and R.Y. designed, fabricated, and characterized the devices. Z.L. and Q.W. designed and fabricated the signal acquisition circuits. Z.L., X.W., W.Z., W.H., and Y.G. analyzed data and prepared figures. Z.L., M.W., and J.L. wrote the manuscript. Y.H., Y.L., M.W. and J.L. supervised this project.

Data availability

The data are available from the corresponding authors upon reasonable requests.

Conflict of interest

The authors declare no competing interests.

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References

- Liu, Y., Pharr, M. & Salvatore, G. A. Lab-on-skin: a review of flexible and stretchable electronics for wearable health monitoring. *ACS Nano* **11**, 9614–9635 (2017).
- Wang, Y. et al. Electrically compensated, tattoo-like electrodes for epidermal electrophysiology at scale. *Sci. Adv.* **6**, eabd0996 (2020).
- Kim, K., Kim, B. & Lee, C. H. Printing flexible and hybrid electronics for human skin and eye-interfaced health monitoring systems. *Adv. Mater.* **32**, 1902051 (2020).
- Wang, Y. et al. Skin bioelectronics towards long-term, continuous health monitoring. *Chem. Soc. Rev.* **51**, 3759–3793 (2022).
- Xu, J. et al. Recent progress and opportunities of wearable non-invasive epidermal sensors for skin disease diagnosis. *Adv. Healthc. Mater.* **14**, 2402891 (2025).
- Xu, J. et al. On-skin epidermal electronics for next-generation health management. *Nano-Micro Lett.* **18**, 25 (2025).
- Koo, J. H. et al. Wearable electrocardiogram monitor using carbon nanotube electronics and color-tunable organic light-emitting diodes. *ACS Nano* **11**, 10032–10041 (2017).
- Park, S. et al. Ultraflexible near-infrared organic photodetectors for conformal photoplethysmogram sensors. *Adv. Mater.* **30**, 1802359 (2018).
- Nawrocki, R. A. et al. Self-adhesive and ultra-conformable, sub-300 nm dry thin-film electrodes for surface monitoring of biopotentials. *Adv. Funct. Mater.* **28**, 1803279 (2018).
- Zhou, W. et al. Gas-permeable, ultrathin, stretchable epidermal electronics with porous electrodes. *ACS Nano* **14**, 5798–5805 (2020).
- Zhao, Y. et al. Ultra-conformal skin electrodes with synergistically enhanced conductivity for long-time and low-motion artifact epidermal electrophysiology. *Nat. Commun.* **12**, 4880 (2021).
- Kim, D. et al. Epidermal electronics. *Science* **333**, 838–843 (2011).
- Wang, S. et al. Mechanics of epidermal electronics. *J. Appl. Mech.* **79**, 031022 (2012).
- Wang, J. et al. Self-supporting electrodes for gas-involved key energy reactions. *Adv. Funct. Mater.* **31**, 2104620 (2021).
- Gao, W. et al. Recent advances in ultrathin materials and their applications in e-skin. *InfoMat* **5**, e12426 (2023).
- Park, J. et al. Conformal fixation strategies and bioadhesives for soft bioelectronics. *Adv. Funct. Mater.* **34**, 2313728 (2024).
- Wang, Y. et al. Low-cost, μm -thick, tape-free electronic tattoo sensors with minimized motion and sweat artifacts. *npj Flex. Electron.* **2**, 1–7 (2018).
- Li, X. et al. Ultracomfortable hierarchical nanonetwork for highly sensitive pressure sensor. *ACS Nano* **14**, 9605–9612 (2020).
- Tian, G. et al. Ultrathin epidermal P(VDF-TrFE) piezoelectric film for wearable electronics. *ACS Appl. Electron. Mater.* **5**, 1730–1737 (2023).
- Li, L. et al. Highly-sensitive ultrathin flexible pressure sensor based on quasi-2D conductive network for human physiological signal monitoring. *Chem. Eng. J.* **498**, 154355 (2024).
- Wong, T. et al. Tattoo-like epidermal electronics as skin sensors for human-machine interfaces. *Soft Sci.* **1**, 10 (2021).
- Meiti, M. A. et al. Transfer printing by kinetic control of adhesion to an elastomeric stamp. *Nat. Mater.* **5**, 33–38 (2006).
- Park, Y. J., Lee, S., Kim, M., Kim, H. & Ahn, J. Graphene-based conformal devices. *ACS Nano* **8**, 7655–7662 (2014).
- Kang, S. et al. Transparent and conductive nanomembranes with orthogonal silver nanowire arrays for skin-attachable loudspeakers and microphones. *Sci. Adv.* **4**, eaas8772 (2018).
- Liu, J. et al. Transfer printing via a PAA sacrificial layer for wrinkle-free PDMS metallization. *J. Mater. Sci-Mater. Electron* **31**, 2347–2352 (2020).
- Konagai, M., Sugimoto, M. & Takahashi, K. High-efficiency GaAs thin-film solar-cells by peeled film technology. *J. Cryst. Growth* **45**, 277–280 (1978).
- Wu, H. et al. Transfer printing of metallic microstructures on adhesion-promoting hydrogel substrates. *Adv. mater.* **27**, 3398–3404 (2015).

28. Fan, F. et al. Damage-free dry transfer printing of ultrathin films with on-demand interfacial adhesion: principles and applications. *Soft Sci.* **5**, 52 (2025).
29. Choi, M. K. et al. Thermally controlled, patterned graphene transfer printing for transparent and wearable electronic/optoelectronic system. *Adv. Funct. Mater.* **25**, 7109–7118 (2015).
30. Guo, R., Sun, X., Yuan, B., Wang, H. & Liu, J. Magnetic liquid metal (Fe-EGal) based multifunctional electronics for remote self-healing materials, degradable electronics, and thermal transfer printing. *Adv. Sci.* **6**, 1901478 (2019).
31. Huang, Y. et al. Direct laser writing-based programmable transfer printing via bioinspired shape memory reversible adhesive. *ACS Appl. Mater. Interfaces* **8**, 35628–35633 (2016).
32. Chen, F. et al. Laser-driven hierarchical “gas-needles” for programmable and high-precision proximity transfer printing of microchips. *Sci. Adv.* **9**, eadk0244 (2023).
33. Shahrjerdi, D. & Bedell, S. W. Extremely flexible nanoscale ultrathin body silicon integrated circuits on plastic. *Nano Lett.* **13**, 315–320 (2013).
34. Blanton, E. W. et al. Spalling-induced liftoff and transfer of electronic films using a van der Waals release layer. *Small* **17**, 2102668 (2021).
35. Zabow, G. Reflow transfer for conformal three-dimensional microprinting. *Science* **378**, 894–898 (2022).
36. Li, A. et al. Drop-printing with dynamic stress release for conformal wrap of bioelectronic interfaces. *Science* **398**, 1127–1132 (2025).
37. Che, L. et al. Soap film transfer printing for ultrathin electronics. *Small* **20**, 2308312 (2024).
38. Che, L. et al. Surface tension-driven transfer printing for ultrathin curved electronics. *Adv. Eng. Mater.* **27**, e2500280 (2025).
39. Hu, X. et al. Nanoengineering ultrathin flexible pressure sensor with superior sensitivity and perfect conformability. *Small* **19**, 2208015 (2023).
40. Kim, H. et al. Skin preparation-free, stretchable microneedle adhesive patches for reliable electrophysiological sensing and exoskeleton robot control. *Sci. Adv.* **10**, eadk5260 (2024).
41. Liu, H. et al. Development of soft dry electrodes: from materials to structure design. *Soft Sci.* **3**, 27 (2023).
42. Olcay, P., Zavanelli, N., Stancil, B. & Majidi, C. Bio-inspired soft electrodes for robust and reusable bioelectronic stickers. *Flex. Print. Electron.* **10**, 035009 (2025).
43. Webb, R. C. et al. Ultrathin conformal devices for precise and continuous thermal characterization of human skin. *Nat. Mater.* **12**, 938–944 (2013).
44. Daus, A. et al. Fast-response flexible temperature sensors with atomically thin molybdenum disulfide. *Nano Lett.* **22**, 6135–6140 (2022).
45. Choong, C. L. et al. Highly stretchable resistive pressure sensors using a conductive elastomeric composite on a micropyramid array. *Adv. Mater.* **26**, 3451–3458 (2014).
46. Huang, Y., Fan, X., Chen, S. C. & Zhao, N. Emerging technologies of flexible pressure sensors: materials, modeling, devices, and manufacturing. *Adv. Funct. Mater.* **29**, 1808509 (2019).
47. Visser, C. W., Amato, D. N., Mueller, J. & Lewis, J. A. Architected polymer foams via direct bubble writing. *Adv. Mater.* **31**, 1904668 (2019).
48. Liu, Q. et al. High-porosity foam-based iontronic pressure sensor with super-high sensitivity of 9280 kPa^{-1} . *Nano-Micro Lett.* **14**, 21 (2022).
49. Jeong, Y., Park, J., Lee, J., Kim, K. & Park, I. Ultrathin, biocompatible, and flexible pressure sensor with a wide pressure range and its biomedical application. *ACS Sens.* **5**, 481–489 (2020).
50. Zhu, Y. et al. Nanoengineering ultrathin flexible pressure sensors with superior sensitivity and wide range via nanocomposite structures. *ACS Sens.* **9**, 4176–4185 (2024).